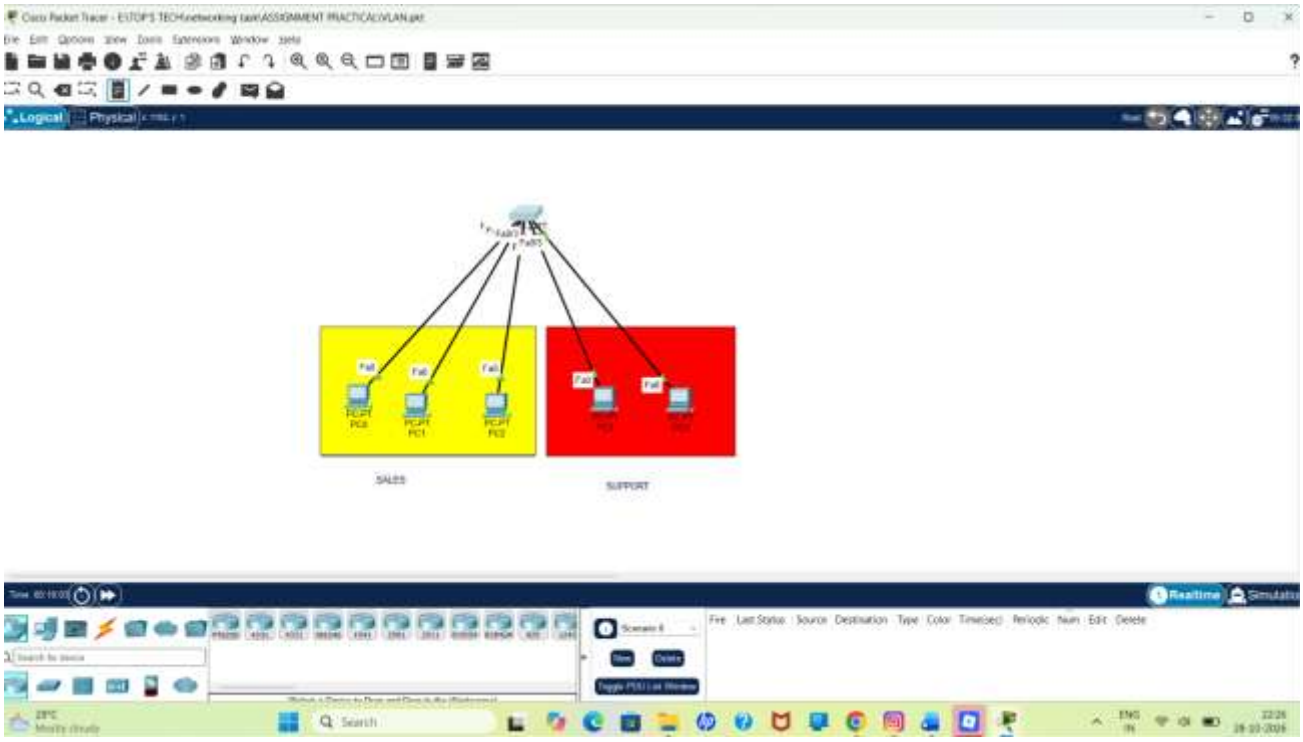
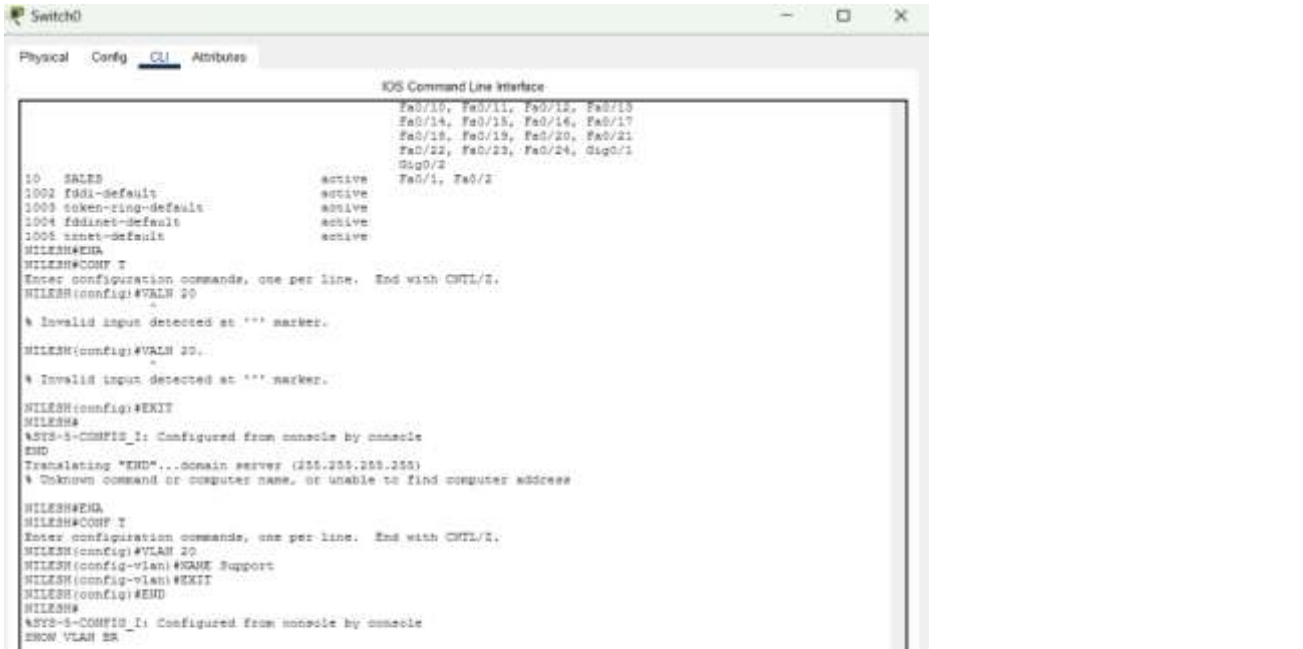
Switch and Router Configuration

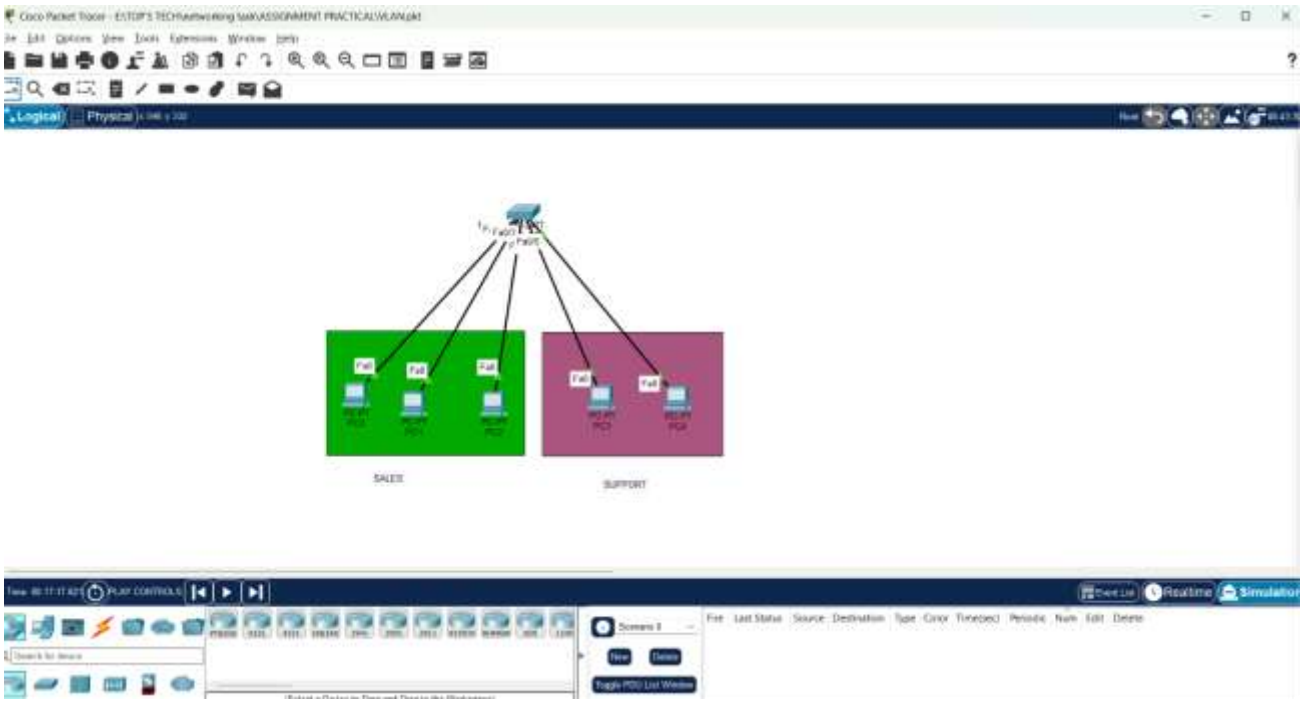
Q1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.Addr

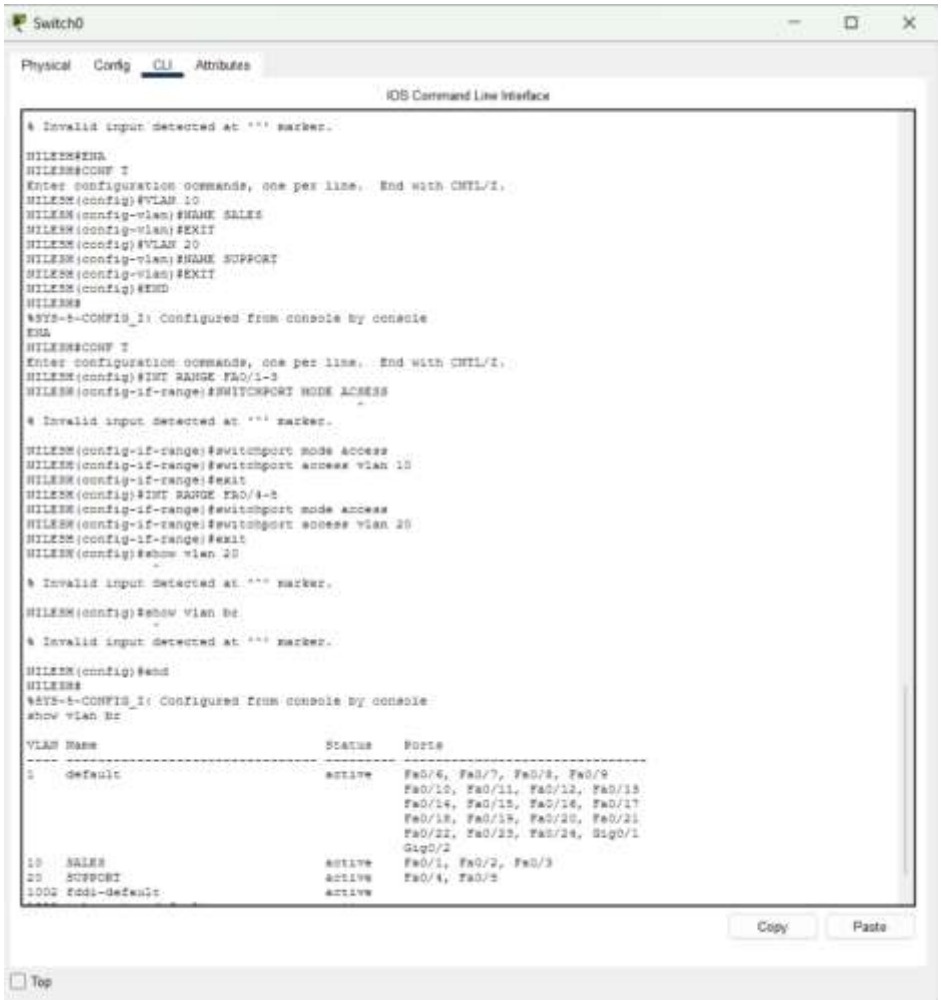




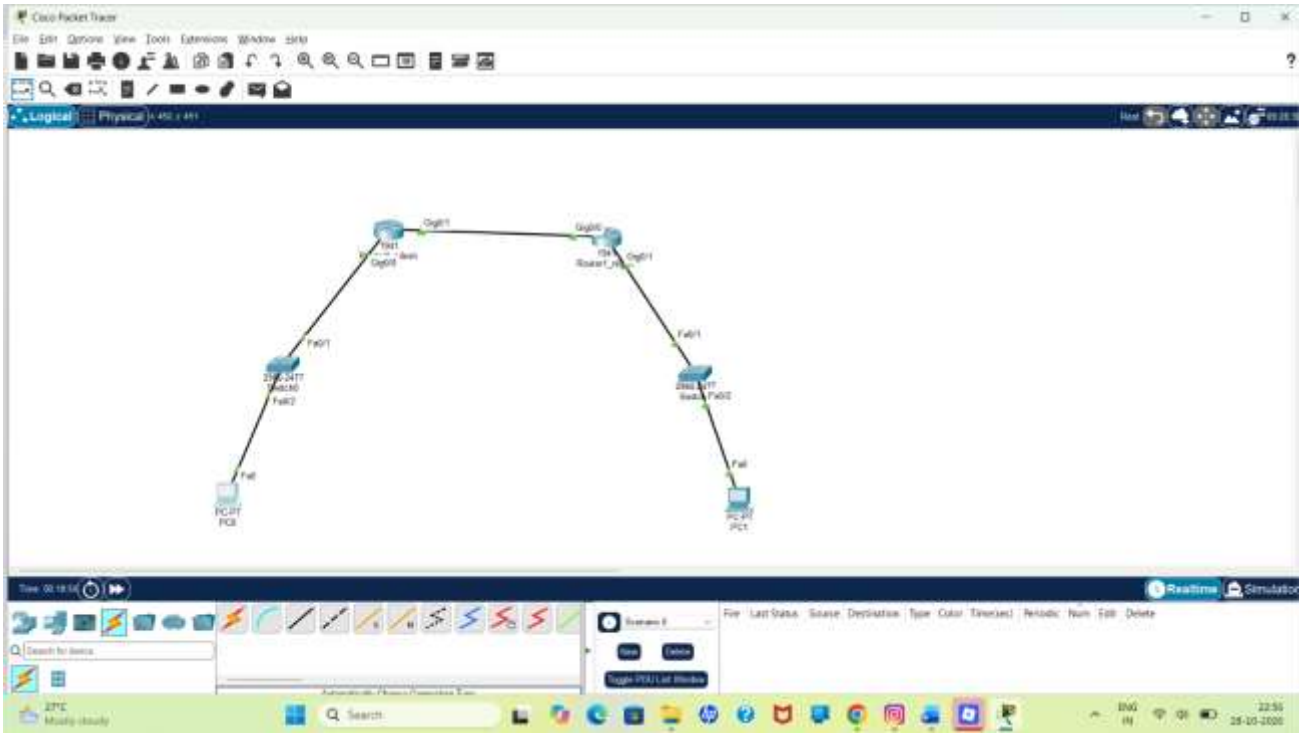
Q2.

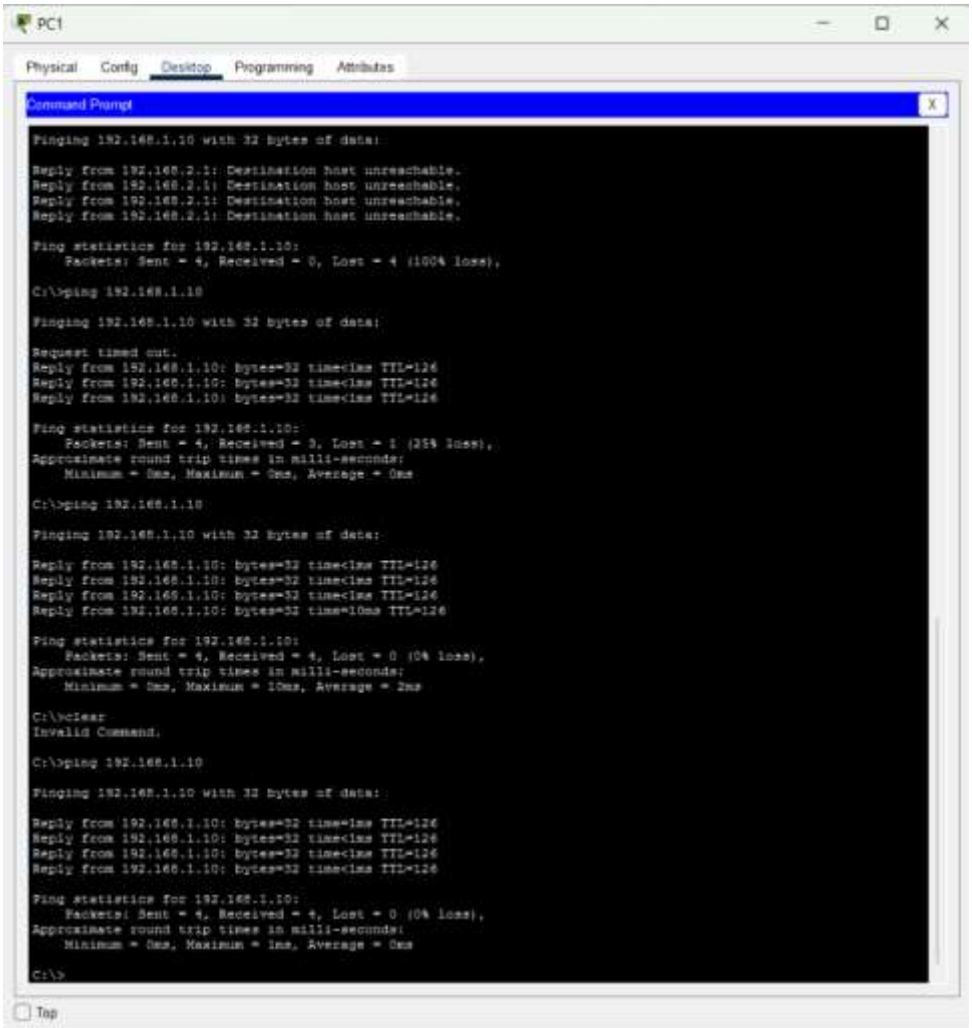
Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.Use the interface range command to select multiple ports efficiently.





Q3 Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.





Q4 Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.Think about router-on-a-stick or Layer 3 switch concepts.

Practical: Inter-VLAN Communication Testing

Aim: To test communication between Sales VLAN 10 and Support VLAN 20 using Cisco Packet Tracer.

Network Configuration

Device	VLAN	IP Address
PC0 (Sales)	VLAN 10	192.168.10.10
PC1 (Support)	VLAN 20	192.168.20.10

Procedure

1. Open Cisco Packet Tracer and add one switch and two PCs.
2. Create VLAN 10 (Sales) and VLAN 20 (Support) on the switch.
3. Assign PC0 to VLAN 10 and PC1 to VLAN 20.
4. Configure the IP addresses on both PCs.
5. Open the Command Prompt on PC0 and execute:

ping 192.168.20.10

Observation

The ping fails because VLAN 10 and VLAN 20 are separate networks. A Layer 2 switch cannot route traffic between different VLANs.

Ping Result: Request timed out (Communication Failed).

Solution

To enable communication between the VLANs, configure **Router-on-a-Stick** or use a **Layer 3 switch**.

Router-on-a-Stick uses router subinterfaces and a trunk connection to route traffic between VLANs.

Example default gateways:

- VLAN 10: 192.168.10.1
- VLAN 20: 192.168.20.1

Conclusion

Devices in different VLANs cannot communicate directly through a Layer 2 switch. Inter-VLAN routing is required to enable successful communication between the Sales and Support VLANs.